

PYTHON WITH AI



Introduction To Python

- Why Python
- Application areas of python
- Python implementations
 1. Cpython
 2. Jython
 3. Ironpython
 4. Pypy
- Python versions
- Installing python
- Python interpreter architecture
- 5. Python byte code compiler
- 6. Python virtual machine(pvm)

Writing and Executing First Python Program

- Using interactive mode
- Using script mode
 1. General text editor and command window
 2. Idle editor and idle shell
- Understanding print() function
- How to compile python program explicitly

Python Language Fundamentals

- Character set
- Keywords
- Comments
- Variables
- Literals
- Operators
- Reading input from console
- Parsing string to int, float

Python Conditional Statements

- If statement
- If else statement
- If elif statement
- If elif else statement
- Nested if statement

Looping Statements

- While loop

- For loop
- Nested loops
- Pass, break and continue keywords

Standard Data Types

- Int, float, complex
- bool, nonetype
- Str, list, tuple, range
- Dict, set, frozenset

String Handling

- What is string
- String representations
- Unicode string
- String functions, methods
- String Repetition and concatenation
- String indexing and slicing
- String formatting

Python List

- Creating and accessing lists
- Indexing and slicing lists
- List methods
- Nested lists
- List comprehension

Python Tuple

- Creating tuple
- Accessing tuple
- Immutability of tuple

Python Set

- How to create a set
- Iteration over sets
- Python set methods
- Python frozenset

Python Dictionary

- Creating a dictionary
- Dictionary methods
- Accessing values from dictionary
- Updating dictionary
- Iterating dictionary
- Dictionary comprehension

Python Functions

- Defining a function
- Calling a function
- Types of functions
- Function v/s Method
- Function arguments

1. Positional arguments, keyword arguments
 2. Default arguments, non-default arguments
 3. Arbitrary arguments, keyword arbitrary arguments
- Function return statement
 - Nested function
 - Function as argument
 - Function as return statement
 - Decorator function
 - Closure
 - Map(), filter(), reduce(), any() functions
 - Anonymous or lambda function

Modules & Packages

- Why modules
- Script v/s module
- Importing module
- Standard v/s third party modules
- Why packages
- Understanding pip utility

File I/O

- Introduction to file handling
- File modes
- Functions and methods related to file handling
- Understanding with block

Object Oriented Programming

- Procedural v/s Object Oriented Programming
- OOP Principles
- Defining a Class & Object Creation
- Inheritance
- Encapsulation
- Polymorphism
- Abstraction
- Garbage Collection
- Iterator & Generator

Exception Handling

- Difference Between Syntax Errors and Exceptions
- Keywords used in Exception Handling
- 1. try , except , finally , raise , assert
- Types of Except Blocks
- User-defined Exceptions

GUI Programming

- Introduction to Tkinter Programming
- Tkinter Widgets
- 1. Tk, Label, Entry, textBox, Buttons
- 2. Frame, messagebox, filedialogetc

- Layout Managers
- Event handling
- Displaying image

Multi-Threading Programming

- Multi-processing v/s Multi-threading
- Need of threads
- Creating child threads
- Functions /methods related to threads
- Thread synchronization and locking

Regular Expressions(Regex)

- Need of regular Expressions
- re module
- Functions /Methods related to regex
- Meta Characters & Special Sequences

SQL

Introduction to RDBMS

- Database Concepts
- What is Relational Database Package
- Difference between SQL & Database
- Installing MySQL Server database

SQL Basic

- DDL: Create, Alter, Drop, etc.
- DML: Insert, Update, Delete ,etc.
- DQL : Select
- Autoincrement field
- SQL Comments
- SQL Aliases
- Savepoint & rollback

SQL Constraints

- Not NULL, Unique key
- Primary key, Check
- Default, Foreign key

SQL Operators

- Arithmetic operators
- Logical operators
- Conditional operators
- Like, between, in operators

SQL Clauses

- Order by
- Where
- Limit/top
- Group by
- having

SQL Joins

- Inner Join
- Left Join
- Right Join
- Full Join

SQL View

- creating view
- updating view
- fetching data from view

SQL Functions

- String functions
 - 1.CHAR_LENGTH
 - 2.CONCAT
 - 3.LOWER
 - 4.REVERSE
 - 5.UPPER
- Aggregate functions
 - 6.MAX, MIN, SUM
 - 7.AVG, COUNT, ABS
- Date & time functions
 - 8.CURDATE
 - 9.CURTIME
 - 10.NOW

Stored Procedures & Functions

- Understanding stored procedures and their key benefits
- Working with stored procedures
- Studying user-defined functions

Working with CSV Files:

- How to write result to csv file
- How to read csv file

Python Database Connectivity

- Database Drivers and connectors
- Creating connection object
- Understanding cursor object
- Executing SQL statements using cursor
- Fetching records from cursor
- Storing and retrieving Date and Time

Statistics & Analytics

Introduction to Statistics

Sample or Populaon

Measures of Central Tendency

- Arithmec Mean
- Harmonic Mean
- Geometric Mean

- Mode
- Quarle
- 1. First quarle
- 2. Second quarle(Median)
- 3. Third quarle
- Standard Deviaon

Probability Distributions

- Introducon to probability
- Condional probability
- Normal Distributon
- Uniform Distributon
- Exponenal Distributon
- Right & Le Skewed Distributon
- Random Distributon
- Central Limit Theorem

Hypothesis Testing

- Normality Test
- Mean Test
- 1. T-test
- 2. Z-test
- 3. ANOVA test
- Chi Square Test
- Correlaon and Covariance

Numpy Package

- Difference between list and numpy array
- Vector and matrix operations
- Array indexing and slicing

Pandas Package

Introdution to pandas

- Labeled and structured data
- Series and dataframe objects
- How to load datasets
- 1. From excel
- 2. From csv
- 3. From html table

How to load datasets

- From excel
- From csv
- From html table

Accessing data from Data Frame

- at & iat
- loc & iloc
- head() & tail()

Exploratory Data Analysis (EDA)

- describe()
- groupby()
- crosstab()
- boolean slicing / query()

Data Manipulation & Cleaning

- Map(), apply()
- Combining data frames
- Adding/removing rows & columns
- 1. Sorting data
- 2. Handling missing values
- 3. Handling duplicacy
- 4. Handling data error
- Categorical Data Encoding
- 5. Label Encoding
- 6. One Hot Encoding
- Handling Date and Time

Data Visualization using matplotlib and seaborn packages

- Scatter plot, lineplot, bar plot
- Histogram, pie chart,
- Jointplot, pairplot, heatmap
- Outlier detection using boxplot

Machine Learning

Introduction To Machine Learning

- Traditional v/s Machine Learning Programming
- Real life examples based on ML
- Steps of ML Programming
- Data Preprocessing revised
- Terminology related to ML

Supervised Learning

- Classification
- Regression

Unsupervised Learning

- clustering

KNN Classification

- Math behind KNN
- KNN implementation
- Understanding hyper parameters

Performance metrics

- Confusion Matrix
- Accuracy Score
- Recall & Precision
- F-1 Score
- R2 Score

Regression

- Math behind Regression
- Simple Linear Regression
- Multiple Linear Regression
- Polynomial Regression
- Boston Price Prediction
- Cost or Loss Functions
- 1. Mean absolute error
- 2. Mean squared error
- 3. Root mean squared error
- 4. Least Square Error
- Regularization

Logistic Regression for classification

- Theory of Logistic Regression
- Binary and Multiclass classification
- Implementing titanic dataset
- Implementing iris dataset
- Sigmoid and softmax functions

Support Vector Machines

- Theory of SVM
- SVM Implementation
- kernel, gamma, alpha

Decision Tree Classification

- Theory of Decision Tree
- Node Splitting
- Implementation with iris dataset
- Visualizing Tree

Ensemble Learning

- Random Forest
- Bagging and Boosting
- Voting Classifier

Model Selection Techniques

- Cross Validation
- Grid and Random Search for hyper parameter tuning

Recommendation System

- Content based technique
- Collaborative filtering technique
- Evaluating similarity based on correlation
- Classification-based recommendations

Clustering

- K-means Clustering
- Hierarchical Clustering
- Elbow technique
- Silhouette coefficient

Text Analysis

- Install NLTK
 - Tokenize words
 - Tokenizing sentences
 - Stop words customization
 - Stemming and Lemmatization
 - Feature Extraction
 - Sentiment Analysis
 - Count Vectorizer
 - TfidfVectorizer
 - Naive Bayes Algorithms
1. GaussianNB
 2. MultinomialNB
 3. BernoulliNB

Dimensionality Reduction

- Principal Component Analysis (PCA)

Open CV

- Reading images
- Understanding Gray Scale Image
- Resizing image
- Understanding Haar Classifiers
- Face,eyes,smile,body classification
- How to use webcam in open cv
- Building image data set
- Capturing video
- Face classification in video

Deep Learning & Neural Networks

Introduction To Artificial Neural Network

- What is Artificial Neural Network (ANN)?
- How Neural Network Works?
- Perceptron
- Multilayer Perceptron
- Feed Forward
- Back propagation

Introduction To Deep Learning

- What is Deep Learning?
 - Deep Learning Packages
 - Deep Learning Applications
 - Building Deep Learning Environment
1. Installing Tensor Flow Locally
 2. Understanding Google Colab

Tensor Flow Basics

- What is Tensorflow?
- Tensorflow 1.x V/S Tensorflow 2.x
- Variables, Constants
- Scalar, Vector, Matrix

- Operations using tensorflow
- Difference between tensorflow and numpy operations
- Computational Graph

Optimizers

- What does optimizers do?
- Gradient Descent (full batch and min batch)
- Stochastic Gradient Descent
- Learning rate, epoch

Activation Functions

- What does Activation Functions do?
- Sigmoid Function,
- Hyperbolic Tangent Function (tanh)
- ReLU -Rectified Linear Unit
- Softmax Function
- Vanishing Gradient Problem

Building Artificial Neural Network

- Using scikit implementation
- Using Tensorflow
- Understanding MNIST Dataset
- Initializing weights and biases
- Gradient Tape
- Defining loss/cost Function
- Train the Neural Network
- Minimizing the loss by adjusting weights and biases

Modern Deep Learning Optimizers and Regularization

- SGD with Momentum
- RMSprop
- AdaGrad
- Adam
- Dropout Layers and Regularization
- Batch Normalization

Building Deep Neural Network Using Keras

- What is Keras?
- Keras Fundamental For Deep Learning
- Keras Sequential Model and Functional API
- Solve a Linear Regression and Classification Problem with Example
- Saving and Loading a Keras Model

Convolutional Neural Networks (CNNs)

- Introduction to CNN
- CNN Architecture
- Convolutional Operations
- Pooling, Stride and Padding Operations
- Data Augmentation
- Building, Training and Evaluating First CNN Model

- Model Performance Optimization
 - Auto encoders for CNN
 - Transfer Learning and Object Detection Using Pre-trained CNN Models
1. LeNet
 2. AlexNet
 3. VGG16
 4. ResNet50
 5. Yolo algorithm

Word Embedding

- What is Word Embedding?
- Word2Vec Embedding
- CBOW
- skipgram
- Keras Embedding Layers
- Visualize Word Embedding
- Google Word2Vec Embedding
- Glove Embedding

Recurrent Neural Networks (RNNs)

- Introduction to RNN
- RNN Architecture
- Types of RNN
- Implementing basic RNN in tensorflow
- Need for LSTM and GRU
- Deep RNN/LSTM/GRU
- Text Classification Using LSTM
- Prediction for Time Series problem
- Bidirectional RNN/LSTM
- Seq-2-Seq Modeling
- Encoder-Decoder Model
- Attention Mechanism

Generative Adversarial Networks (GANs)

- Introduction to GAN
- Generator
- Discriminator
- Types of GAN
- Implementing GAN using Neural Network

Speech Recognition APIs

- Text To Speech
- Speech To Text
- Automate task using voice
- Voice Search on Web

Integration of ChatGPT API with Python

- Introduction to ChatGPT
- Understanding openai library
- Registering for an API key
- API documentation and resources
- Type of ChatGPT Models
- Generating Images from ChatGPT API
- Image Captioning using ChatGPT API
- Building a Chatbot with ChatGPT API and Python

Projects(Any Five)

- Stock Price Prediction Using LSTM
- Object Detection
- Attendance System Using Face Recognition
- Facial Expression and Age Prediction
- Chabot Application
- Neural Machine Translation
- Hand Written Digits& Letters Prediction
- Number Plate Recognition
- Gender Classification
- My Assistant for Desktop
- Suspect Detection using CCTV
- Hardware operations using gesture detection
- Cat v/s Dog Image Classification



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